B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR SECOND SEMESTER EXAMINATION, 2019
[According to the New Syllabus]
CSE-510225 (Linear Algebra)
Time 3 hours
Full marks 80

N.B. The figures in the right margin indicate full marks. Answer any four questions.

1. (a) Define linear equation with examples.

(b) Determine the value of λ such that the following system of linear equations in unknowns x,y, z has:

(i) a unique solution
(ii) no solution
(iii) more than one solution.

(c) Define norm of a vector. Consider the points, P (3, λ, -2) and Q (5, 3, 4) in R^3. Find the value of λ so that $\overrightarrow{PQ}$ is orthogonal to the vector (4, -3,2).

(d) Prove that $\begin{bmatrix} -a & b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & -c & b & a \end{bmatrix} = (a^2 + b^2 + c^2 + d^2)^2$

2. (a) Define: (I) symmetric matrix
(ii) square matrix
(iii) idempotent matrix
(iv) singular matrix

(b) If A and B are comparable matrices and A^T and B^T are the transpose matrices of A and B respectively, then prove that (i) $(A^T)^T = A$, (ii) $(A+B)^T = A^T + B^T$, (iii) $(AB)^T = (B^T A^T)$

(c) Solve the following system of linear equations with the help of matrix:

$x + 2y + 3z + 4 = 0$
$2x + 4y + 5z + 7 = 0$
$3x + 5y + 6z + 10 = 0$

(d) Show that the matrix A = is $\begin{vmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ 4 & 4 & -3 \end{vmatrix}$ idempotent.

3. (a) Define linear transformation. Show that the product of two linear transformation is a linear transformation.

   (b) Let S and T be the linear operators of $\mathbb{R}^2$ into $\mathbb{R}^2$ defined by $S(x, y) = 3x - 2y, - 6x + y)$ $T(x, y) = (2x + y, x - y)$. Find formulae defining properties $S + T$, ST, TS and S^2.

   (c) Define (i) System of linear equation
   (ii) Consistent
   (iii) Homogeneous
   (iv) Non-homogeneous

4. (a) Define minors and co-factors with example.

   (b) If $\Delta = \begin{vmatrix} x & x^2 & x^3+1 \\ y & y^2 & y^3-1 \\ z & z^2 & z^3-1 \end{vmatrix}$ then prove that,

   $\Delta = (x-y)(y-z)(z-x)(xyz-1)$. Also show that if x, y and z are not equal and $\Delta = 0$, then $xyz + 1 = 0$.

   (c) Prove that,

   $$\begin{vmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & d & d^2 & d^3 \end{vmatrix} = (d-c)(d-b)(d-a)(c-b)(c-a)(b-a).$$

   (d) Prove that, $\begin{vmatrix} 1+a_1 & a_2 & a_3 & a_4 \\ a_1 & 1+a_2 & a_3 & a_4 \\ a_1 & a_2 & 1+a_3 & a_4 \\ a_1 & a_2 & a_3 & 1+a_4 \end{vmatrix} = 1+a_1+a_2+a_3+a_4.$

5. (a) Define image and kernel of a linear transformation.

   (b) Show that, $T: \mathbb{R}^3 \to \mathbb{R}^2$, where $T(x, y, z) = (x+y-z, 2x-y+2z)$ is a linear transformation. Find a basis and dimension for ImT and ker T.

   (c) Find a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^4$ whose ImT is generated by $\{(1, 2, 0, -4), (2, 0, -1, -3)\}$.

6. (a) Define characteristic matrix and characteristic equation.

   (b) Find the eigen values and eigen vectors of the matrix

   $$A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$$

   (c) State Cayley-Hamilton theorem. Using this theorem find the inverse of the matrix $A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.